Weil Conjectures

Chapter 19: Proof of the Weil conjectures

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19 Proof of the Weil conjectures

The Deligne–Laumon theorem is an upper bound for weights, stated with ℓ-adic coefficients and one embedding $\iota : \overline{\mathbf{Q}}_\ell \to \mathbf{C}$. Weil's conjectures ask for more: an equality $|\alpha| = q^{i/2}$, for polynomials P_i with integer coefficients, under every complex embedding. This chapter closes both gaps.

The lower bound comes from Poincaré duality, and a curve shows how. Let X_0 be a smooth projective curve and α an eigenvalue of Frobenius on $H^1(X, \overline{\mathbf{Q}}_\ell)$. Then Theorem 13.11*, applied to the constant sheaf, gives $|\iota(\alpha)| \leq q^{1/2}$. On the other hand, cup product is a perfect pairing $H^1 \times H^1 \to H^2 = \overline{\mathbf{Q}}_\ell(-1)$, which Frobenius multiplies by q. Hence q/α is an eigenvalue as well, and the same bound for q/α says that $|\iota(\alpha)| \geq q^{1/2}$. In dimension d the argument is the same: an eigenvalue α on $H^i(X, \overline{\mathbf{Q}}_\ell)$ satisfies $|\iota(\alpha)| \leq q^{i/2}$, while duality makes q^d/α an eigenvalue on H^{2d-i}, of absolute value at most $q^{(2d-i)/2}$, and we find $|\iota(\alpha)| = q^{i/2}$. This equality is Weil's axiom (d), and we prove it in Section 19.1, more generally for the higher direct images of a pure lisse sheaf under a smooth proper morphism (Theorem 19.1); the case of the constant sheaf over a point is Corollary 19.2. Since the Deligne–Laumon theorem holds for every ι, purity holds for all complex embeddings at once. Indeed, every embedding of a number field into $\mathbf{C}$ extends to an isomorphism $\overline{\mathbf{Q}}_\ell \simeq \mathbf{C}$.

The polynomials $P_{i,\ell} = \det(1 - TF \mid H^i(X, \mathbf{Q}_\ell))$ are defined with ℓ-adic coefficients, and a priori they could depend on ℓ. The zeta function does not mention ℓ. It has a unique expression $Z(X_0, T) = A/B$ with $A, B \in \mathbf{Z}[T]$ coprime and $A(0) = B(0) = 1$ (Lemma 19.4; Sections 19.2–Section 19.3). Can each $P_{i,\ell}$ be read off from $Z(X_0, T)$? The trace formula writes Z as the alternating product of the $P_{i,\ell}$, but an alternating product can cancel. For example, $X_0 = \mathbf{G}_m \sqcup \operatorname{Spec} \mathbf{F}_q$ is not proper, and it has q^m points over $\mathbf{F}_{q^m}$, as many as $\mathbf{A}^1$. Hence $Z(X_0, T) = 1/(1 - qT)$. Yet H^0_c and H^1_c are both one-dimensional with eigenvalue 1, and the factor $1 - T$ cancels between them. For a smooth proper X_0 purity rules this out. The reciprocal roots of $P_{i,\ell}$ have absolute value $q^{i/2}$, so no root of $P_{i,\ell}$ is a root of $P_{j,\ell}$ for $i \neq j$. It follows that the product of the $P_{i,\ell}$ with i odd is A and the product with i even is B. Grouping the reciprocal

roots of A and B by absolute value then recovers each $P_{i,\ell}$ from $Z(X_0, T)$ alone (Theorem 19.7). In particular $P_{i,\ell}$ has integer coefficients and does not depend on ℓ.

In terms of point counts, purity gives the classical estimates. For a smooth proper geometrically connected X_0 we obtain $|\#X_0(\mathbf{F}_{q^m}) - (1+q^{dm})| \leq \sum_{i=1}^{2d-1} b_i q^{mi/2}$ (Corollary 19.10). In particular, for a curve of genus g this is Weil's bound $2gq^{m/2}$, and for elliptic curves it is Hasse's theorem. As an illustration, Example 19.9 works this out for genus-one curves over $\mathbf{F}_3$.

19.1 The Riemann hypothesis

For a proper map we have $Rf_! = Rf_*$, so the upper bound of Theorem 18.15* applies to ordinary cohomology. If the map is also smooth, Poincaré duality on each fibre pairs $H^i(X_{\bar{y}}, F)$ perfectly with $H^{2d-i}(X_{\bar{y}}, F^\vee(d))$. The eigenvalues of the second group are the inverses of those of the first. The second group has weights at most $(-w-2d)+(2d-i) = -w-i$, and this upper bound becomes the lower bound $w+i$ for the first.

Both hypotheses are needed. Without properness, H^1_c of $\mathbf{G}_m$ has weight 0 instead of 1. Without smoothness duality fails. For example, the nodal cubic is proper, and its H^1 is a line on which Frobenius acts by ± 1, of weight 0. The relative statement is no harder than the absolute one, because we check the bound at one closed point y at a time, on the smooth proper fibre over $\kappa(y)$. In the proof of Theorem 19.7 we use only the case of a point as base and the constant sheaf (Corollary 19.2).

Theorem 19.1 (Purity of higher direct images under a smooth proper morphism)**.** *Let $f_0 : X_0 \to Y_0$ be smooth proper of pure relative dimension d, and let F be a lisse Weil sheaf, punctually ι-pure of weight w. For every i, the canonical identification*

$$R^i f_{0!} F = R^i f_{0*} F$$

is Weil-equivariant, and this constructible sheaf is punctually ι-pure of weight $w+i$. In particular, it is mixed of the single weight $w+i$.

Proof. Since f_0 is proper, the definition of $Rf_{0!}$ by compactification identifies it with Rf_{0*}. This identification commutes with the base change maps that define the Weil structures, which proves the first assertion.

Fix a closed point y. By proper base change, the stalk at $\bar{y}$ with its local Frobenius is $H^i(X_{\bar{y}}, F)$. By Theorem 18.15*, $R^i f_{0*} F$ is mixed of weights at most $w+i$, so every eigenvalue α there satisfies

$$|\iota(\alpha)| \leq q_y^{(w+i)/2}.$$

The opposite inequality comes from duality. The dual $F^\vee$ is lisse of weight $-w$, so $F^\vee(d)$ has weight $-w-2d$. On the smooth proper geometric fibre, Corollary 10.13* and Theorem 10.21* give a Frobenius-equivariant perfect pairing

$$H^i(X_{\bar{y}}, F) \times H^{2d-i}(X_{\bar{y}}, F^\vee(d)) \longrightarrow \overline{\mathbf{Q}}_\ell.$$

Since the pairing is perfect and takes values in the trivial Weil line, it identifies the second factor with the dual of the first as a Frobenius module, so the eigenvalues on the second factor are the α^{-1}, with multiplicities. Applying Theorem 18.15* once more, now to the structure map of the fibre over $\kappa(y)$ and to $F^\vee(d)$, of weight $-w-2d$, and to its cohomology in degree $2d-i$, we get

$$|\iota(\alpha)|^{-1} \le q_y^{(-w-2d+2d-i)/2} = q_y^{(-w-i)/2}.$$

The two inequalities give $|\iota(\alpha)| = q_y^{(w+i)/2}$ at every closed point y. Every nonzero pure quotient in a mixed filtration of $R^i f_{0*} F$ has a nonzero stalk at some closed point, so its weight is $w+i$. This proves the last assertion. ∎

For the Weil conjectures we need only the case where the base is a point and the sheaf is constant. Then the fibre is X itself, and the stalk of $R^i f_* \overline{\mathbf{Q}}_\ell$ is $H^i(X, \overline{\mathbf{Q}}_\ell)$.

Corollary 19.2 (Purity of the cohomology of a smooth proper variety)**.** *Let $X_0/\mathbf{F}_q$ be smooth proper of pure dimension d, put $X = X_0 \times_{\mathbf{F}_q} \overline{\mathbf{F}}_q$, let $\ell \nmid q$ be a prime, and let $\iota : \overline{\mathbf{Q}}_\ell \hookrightarrow \mathbf{C}$ be a field embedding. Then every geometric Frobenius eigenvalue α on $H^i(X_{\text{proét}}, \overline{\mathbf{Q}}_\ell)$ satisfies*

$$|\iota(\alpha)| = q^{i/2}.$$

Proof. The constant sheaf has weight zero, since its local Frobenius is the identity. So Theorem 19.1 applies to the structure map $f_0 : X_0 \to \operatorname{Spec} \mathbf{F}_q$ with $w = 0$. The base has a single closed point, with $q_y = q$. The stalk of $R^i f_{0*} \overline{\mathbf{Q}}_\ell$ there is $H^i(X, \overline{\mathbf{Q}}_\ell)$ with its geometric Frobenius, and its eigenvalues therefore satisfy $|\iota(\alpha)| = q^{i/2}$. ∎

19.2 Rational power series

The zeta function has integer coefficients by its definition, but the trace formula proves its rationality only over $\mathbf{Q}_\ell$, as an alternating product of ℓ-adic characteristic polynomials. To compare these polynomials for different primes ℓ, we need a form of the zeta function that does not mention ℓ. We show that a rational power series with integer coefficients is a quotient of integral polynomials in a unique reduced form.

Lemma 19.3 shows that rationality is a condition of linear algebra on the coefficients, namely that the Hankel matrix (c_{r+s}) has finite rank. Such a condition descends from $\mathbf{Q}_\ell$ to $\mathbf{Q}$. For integrality of the reduced fraction, Lemma 19.4 uses a lattice in the row span of the Hankel matrix that is stable under the shift. The characteristic polynomial of the shift on this lattice is then integral. For example, the coefficients $c_n = 2^{n+1} - 1$ of $1/((1-2T)(1-T))$ satisfy $c_{n+2} = 3c_{n+1} - 2c_n$. The shift matrix S on the row span has $\det(1 - TS) = (1-2T)(1-T)$, which is the reduced denominator. In the next section we compare the ℓ-adic factorizations with the reduced fraction A/B.

Lemma 19.3 (Hankel rank and rationality)**.** *For a field K and $C(T) = \sum_{n \ge 0} c_n T^n \in K[[T]]$, the following conditions are equivalent:*

1. *$C(T)$ is rational and regular at zero;*
2. *the finite submatrices of $(c_{r+s})_{r,s \ge 0}$ have bounded rank;*
3. *the span of the rows $R_j = (c_{j+s})_{s \ge 0}$ is finite-dimensional.*

If $K \subset K'$ and C has coefficients in K, rationality over K' implies rationality over K.

Proof. The equivalence of (2) and (3) is linear algebra. For the equivalence of (1) and (3), note that a denominator Q gives a linear recurrence among the rows. Conversely, the characteristic polynomial of the shift on a finite-dimensional row span is a denominator. The descent statement holds because (2) does not change when we extend the field.

For example, $C = 1/(1-T)$ has all $c_n = 1$, every row is $(1, 1, \ldots)$, and the row span is a line. For $C = 1/(1-T)^2$ we have $c_n = n+1$, and $R_j = R_0 + j(R_1 - R_0)$, so the row span is a plane.

For (3)⇒(2), a nonzero $k \times k$ minor of a finite submatrix gives k linearly independent rows. So every finite submatrix has rank at most the dimension of the row span. For (2)⇒(3), let $R_{j_1}, \ldots, R_{j_m}$ be linearly independent. On their span the coordinate functionals have common kernel zero. They therefore span its dual, and m of them already do. The corresponding $m \times m$ minor is nonzero, and so m is at most the bound in (2).

For (1)⇒(3), suppose that $QC = P$ with $Q(0) = 1$. Comparing coefficients gives a fixed linear recurrence for all sufficiently large indices, and the same recurrence expresses every sufficiently late row in terms of a fixed finite list of earlier rows.

For (3)⇒(1), the left shift $S(v)_n = v_{n+1}$ preserves the finite-dimensional row span V, since $S(R_j) = R_{j+1}$. Let e be evaluation at coordinate zero and $v = R_0$. In any basis of V, with M the matrix of the shift, we have

$$C(T) = e(I - TM)^{-1}v = \frac{e\,\mathrm{adj}(I - TM)v}{\det(I - TM)}.$$

The first equality holds coefficient by coefficient. Indeed, $(I - TM)^{-1} = \sum_n T^n M^n$, and $eM^n v$ is the coordinate zero of $S^n(R_0) = R_n$, that is, c_n. As the denominator has constant coefficient one, this proves (1).

For the descent, note that condition (2) involves only the coefficients $c_n \in K$, and the rank of a matrix with entries in K does not change when we pass to K'. So (2) holds over K if and only if it holds over K'. Applying (1)⇒(2) over K' and (2)⇒(1) over K gives the descent of rationality. ∎

Applied with $K = \mathbf{Q}$ and $K' = \mathbf{Q}_\ell$, the lemma shows that the zeta function, which the trace formula writes as a quotient of polynomials over $\mathbf{Q}_\ell$, is a quotient of polynomials over $\mathbf{Q}$. The next lemma shows that the integrality of its coefficients passes to the reduced fraction.

Lemma 19.4 (Reduced fractions with integer coefficients)**.** *If $C(T) \in 1 + T\mathbf{Z}[[T]]$ is rational over $\mathbf{Q}$, then there are unique coprime $A, B \in \mathbf{Z}[T]$ such that*

$$C(T) = A(T)/B(T), \qquad A(0) = B(0) = 1.$$

Proof. We repeat the construction in the proof of Lemma 19.3 with an integral basis of the row span. This gives an integral fraction, which we then reduce with Gauss' lemma.

Let V be the rational row span of Lemma 19.3, and put $L = V \cap \mathbf{Z}^{\mathbf{Z}_{\geq 0}}$. Then L contains all rows, spans V, and is preserved by the left shift. Since V is finite-dimensional, some r of the coordinate functionals span its dual, as in the proof of (2)⇒(3) in Lemma 19.3, and so they are jointly injective on V. These coordinates embed L into $\mathbf{Z}^r$. Since a subgroup of $\mathbf{Z}^r$ is free of finite rank, L is a shift-stable lattice in V, free of rank $\dim V$.

In a basis of L, the shift matrix M, the row e and the column $v = R_0$ are integral. The matrix identity in the proof of Lemma 19.3 yields integral polynomials

$$A_0 = e\,\mathrm{adj}(I - TM)v, \qquad B_0 = \det(I - TM), \qquad C = A_0/B_0.$$

Both constant coefficients are one. Let G be a primitive integral polynomial that is a greatest common divisor of A_0 and B_0 in $\mathbf{Q}[T]$. By Gauss' lemma the quotients A_0/G and B_0/G are

integral. Since $G(0)$ divides $B_0(0) = 1$, we may choose the sign of G so that $G(0) = 1$. Dividing by G, we obtain polynomials A and B that are coprime and have constant coefficients one.

For uniqueness, let $A/B = A'/B'$ be two such fractions, so that $AB' = A'B$ in $\mathbf{Q}[T]$. Since B is coprime to A, it divides B', and in the same way B' divides B. So $B' = cB$ for a scalar c. Comparing constant coefficients gives $c = 1$, and then $AB = A'B$ forces $A = A'$. ∎

By Corollary 12.31*, $Z(X_0, T) \in \mathbf{Q}(T) \cap (1 + T\mathbf{Z}[[T]])$ for every separated X_0 of finite type over $\mathbf{F}_q$. So Lemma 19.4 applies. We write $Z(X_0, T) = A/B$ for the resulting fraction, with A and B coprime in $\mathbf{Z}[T]$ and $A(0) = B(0) = 1$.

19.3 Integrality and independence of ℓ

Let X_0 be smooth and proper over $\mathbf{F}_q$, of pure dimension d, and let A/B be the reduced fraction of $Z(X_0, T)$ (Lemma 19.4). We now carry out the argument sketched at the start of the chapter. The point is that purity makes the characteristic polynomials of Frobenius in different degrees pairwise coprime, so no cancellation can occur, and the polynomials can be read off from A and B. For a prime $\ell \nmid q$ put

$$H^i_\ell = H^i(X_{\mathrm{pro\acute{e}t}}, \mathbf{Q}_\ell), \qquad P_{i,\ell}(T) = \det(1 - TF \mid H^i_\ell), \qquad b_{i,\ell} = \dim_{\mathbf{Q}_\ell} H^i_\ell.$$

Since X_0 is proper, $H^i_c = H^i$, and the determinant identity of Corollary 12.31* for $\Lambda = \mathbf{Q}_\ell$ reads

$$Z(X_0, T) = \prod_{i=0}^{2d} P_{i,\ell}(T)^{(-1)^{i+1}}. \tag{19.1}$$

Since Frobenius is invertible, $\deg P_{i,\ell} = b_{i,\ell}$.

Definition 19.5 (Reciprocal roots)**.** A *reciprocal root* of a polynomial P over a field is an element α of an algebraic closure such that $1 - \alpha T$ divides P.

Thus the reciprocal roots of $P_{i,\ell}$ are the Frobenius eigenvalues on H^i_ℓ, counted with multiplicity. The next lemma shows that A and B sort these eigenvalues by degree, independently of ℓ and of the complex embedding.

Lemma 19.6 (Reciprocal roots of the zeta function)**.** *Let $X_0/\mathbf{F}_q$ be smooth proper of pure dimension d, let $Z(X_0, T) = A/B$ be the reduced fraction fixed above, and let $\ell \nmid q$ be a prime. Then*

$$A = \prod_{i \text{ odd}} P_{i,\ell}, \qquad B = \prod_{i \text{ even}} P_{i,\ell} \quad \text{in } \mathbf{Q}_\ell[T]. \tag{19.2}$$

Every reciprocal root $a \in \overline{\mathbf{Q}}$ of A or of B is a nonzero algebraic integer, and there is an integer i, odd for A and even for B, such that $|\tau(a)| = q^{i/2}$ for every field embedding $\tau : \mathbf{Q}(a) \hookrightarrow \mathbf{C}$ and such that $\theta(a)$ is a Frobenius eigenvalue on H^i_ℓ for every field embedding $\theta : \overline{\mathbf{Q}} \hookrightarrow \overline{\mathbf{Q}}_\ell$.

Proof. The identity (19.2) holds because both sides are reduced fractions for $Z(X_0, T)$ over $\mathbf{Q}_\ell$. By Corollary 19.2, the reciprocal roots of $P_{i,\ell}$ have absolute value $q^{i/2}$ under every embedding $\iota : \overline{\mathbf{Q}}_\ell \hookrightarrow \mathbf{C}$. Since $q > 1$, distinct degrees give distinct absolute values, and so the polynomials $P_{i,\ell}$ are pairwise coprime. Their products over odd and over even degrees are therefore coprime, and they have constant coefficients one. By (19.1) their quotient is $Z(X_0, T)$. On the other

hand, A and B remain coprime in $\mathbf{Q}_\ell[T]$, since a Bézout relation over $\mathbf{Q}$ remains valid over $\mathbf{Q}_\ell$. The two reduced fractions agree by the uniqueness argument in the proof of Lemma 19.4, which works over any field.

Write $A(T) = \prod_a (1 - aT)$ and $B(T) = \prod_b (1 - bT)$ over $\overline{\mathbf{Q}}$. The reciprocal roots a and b are roots of the monic integral polynomials $U^{\deg A} A(U^{-1})$ and $U^{\deg B} B(U^{-1})$. Hence they are nonzero algebraic integers. Let $\theta : \overline{\mathbf{Q}} \hookrightarrow \overline{\mathbf{Q}}_\ell$ be an embedding, and let a be a reciprocal root of A or B. By (19.2) and the pairwise coprimality of the $P_{i,\ell}$, there is exactly one i such that $\theta(a)$ is a Frobenius eigenvalue on H^i_ℓ. It is odd for A and even for B.

Corollary 19.2 bounds $|\iota(\theta(a))|$ for embeddings ι of $\overline{\mathbf{Q}}_\ell$. The lemma, however, concerns the embeddings of the number field $\mathbf{Q}(a)$. We pass from one to the other by extending field embeddings. Let $\tau : \mathbf{Q}(a) \hookrightarrow \mathbf{C}$ be any embedding. The map $\tau \circ \theta^{-1}$, defined on $\theta(\mathbf{Q}(a))$, extends to a field isomorphism $\iota' : \overline{\mathbf{Q}}_\ell \xrightarrow{\sim} \mathbf{C}$, because both are algebraically closed fields of characteristic zero with transcendence degree $2^{\aleph_0}$ over $\mathbf{Q}$. We have $\iota'(\theta(a)) = \tau(a)$. Applying Corollary 19.2 with this ι' to the eigenvalue $\theta(a)$ on H^i_ℓ, we find

$$|\tau(a)| = q^{i/2}.$$

Since $q > 1$, this absolute value determines i, and so i does not depend on θ. ∎

Theorem 19.7 (The Weil conjectures for smooth proper schemes)**.** *Let $X_0/\mathbf{F}_q$ be smooth proper of pure dimension d, and let $P_{i,\ell}$ be as above.*

1. *There are unique polynomials $P_i(T) \in \mathbf{Z}[T]$ with $P_i(0) = 1$ whose images in $\mathbf{Q}_\ell[T]$ are $P_{i,\ell}(T)$ for every prime $\ell \nmid q$, and $Z(X_0, T) = \prod_{i=0}^{2d} P_i(T)^{(-1)^{i+1}}$.*
2. *The reciprocal roots of P_i are algebraic integers, and each of them has absolute value $q^{i/2}$ under every complex embedding.*
3. *The degrees $b_i = \deg P_i = \dim_{\mathbf{Q}_\ell} H^i_\ell$ do not depend on ℓ, and $b_i = b_{2d-i}$. If X_0 is geometrically connected, then $P_0 = 1 - T$, $P_{2d} = 1 - q^d T$, and $b_0 = b_{2d} = 1$.*
4. *Put $\chi = \sum_i (-1)^i b_i$. Then $d\chi$ is even and there is $\varepsilon \in \{1, -1\}$ such that*

$$Z\left(X_0, \frac{1}{q^d T}\right) = \varepsilon q^{d\chi/2} T^\chi Z(X_0, T) \quad \text{in } \mathbf{Q}(T).$$

Proof. For (1) and (2), we define P_i using only the reduced fraction A/B of Lemma 19.4, by sorting the reciprocal roots of A and B by absolute value, and then we compare P_i with $P_{i,\ell}$ by Lemma 19.6.

For odd i, let R_i be the submultiset of the reciprocal roots of A whose absolute value under every complex embedding is $q^{i/2}$, and for even i define R_i in the same way using the reciprocal roots of B. By Lemma 19.6, every reciprocal root of A or B lies in exactly one R_i. The multisets of reciprocal roots of A and B are Galois-stable, with multiplicities. The condition on all complex embeddings is Galois-invariant, and so each R_i is Galois-stable. Put

$$P_i(T) = \prod_{a \in R_i} (1 - aT).$$

The coefficients of P_i are integers, being rational by Galois invariance and algebraic integers because the roots satisfy monic integral equations.

Now let $\ell \nmid q$ be any prime and $\theta : \overline{\mathbf{Q}} \hookrightarrow \overline{\mathbf{Q}}_\ell$ an embedding. By (19.2) and Lemma 19.6, the embedding θ maps R_i onto the multiset of Frobenius eigenvalues on H^i_ℓ. Since both polynomials

have constant coefficient one, $P_i = P_{i,\ell}$ in $\overline{\mathbf{Q}}_\ell[T]$, hence in $\mathbf{Q}_\ell[T]$. As ℓ was arbitrary, this proves the existence in (1), and uniqueness follows by comparing with a single $P_{i,\ell}$. The product formula for $Z(X_0, T)$ in (1) is then (19.1), and assertion (2) is Lemma 19.6.

The remaining assertions follow from Poincaré duality and the functional equation of Chapter 12*. For (3), since $P_i = P_{i,\ell}$, we have $b_i = \deg P_i = \dim_{\mathbf{Q}_\ell} H^i_\ell$, which does not depend on ℓ, and $b_i = b_{2d-i}$ by Corollary 12.33*.

For the second part of (3), assume that X_0 is geometrically connected, so that X is connected. By Theorem 10.15*, $H^0(X, \mathbf{Q}_\ell) = \mathbf{Q}_\ell$, with Frobenius acting as the identity, and the trace map tr_X identifies $H^{2d}(X, \mathbf{Q}_\ell)$ with $\mathbf{Q}_\ell(-d)$, on which Frobenius acts by q^d. So $b_0 = b_{2d} = 1$, $P_0 = 1 - T$ and $P_{2d} = 1 - q^d T$.

For (4), Poincaré duality pairs the eigenvalues on H^i_ℓ with those on H^{2d-i}_ℓ by $\alpha \mapsto q^d/\alpha$. Substituting $T \mapsto 1/(q^d T)$ in the product formula therefore exchanges the corresponding factors up to a monomial, and Corollary 12.33* carries out this computation and gives (4) in $\mathbf{Q}_\ell(T)$. Since the zeta function is rational over $\mathbf{Q}$, and the scalar $\varepsilon q^{d\chi/2}$ is a rational number, the identity holds in $\mathbf{Q}(T)$. ■

In the following examples we read the theorem backwards. For projective space the point count determines Z, and the grouping of the reciprocal roots by absolute value recovers the P_i. For an elliptic curve the theorem turns one point count into an inequality.

Example 19.8 (Projective space and elliptic curves)**.** We have $\#\mathbf{P}^d(\mathbf{F}_{q^m}) = 1 + q^m + \cdots + q^{dm}$, so that $Z(\mathbf{P}^d_{\mathbf{F}_q}, T) = \prod_{i=0}^d (1 - q^i T)^{-1}$. Grouping the reciprocal roots by absolute value as in (2), we find $P_{2i} = 1 - q^i T$ for $0 \le i \le d$ and $P_i = 1$ for odd i. For an elliptic curve C_0, we have $P_0 = 1 - T$ and $P_2 = 1 - qT$ by (3). Since $b_1 = 2$ (Appendix D*), the polynomial P_1 has two reciprocal roots α, β. Poincaré duality matches them as $\beta = q/\alpha$, so $P_1 = 1 - aT + qT^2$ with $a = \alpha + \beta$. The point-counting formula for $m = 1$ gives $\#C_0(\mathbf{F}_q) = 1 - a + q$, that is, $a = 1 + q - \#C_0(\mathbf{F}_q) \in \mathbf{Z}$. By (2) both α and β have absolute value $q^{1/2}$, and we get Hasse's theorem $|a| \le 2q^{1/2}$.

Example 19.9 (The running example: purity)**.** For the curve $C_0 \colon y^2 = x^3 - x$ over $\mathbf{F}_3$ of the introduction, the reciprocal roots $\pm i\sqrt{3}$ of $P_1 = 1 + 3T^2$ (Example 12.32*) have absolute value $\sqrt{3}$, as (2) predicts. What purity adds to the earlier examples is a bound: for every genus-one curve over $\mathbf{F}_3$ we get $|4 - N_1| \le 2\sqrt{3}$, that is, $1 \le N_1 \le 7$. In the proof, the upper bound $|\alpha| \le \sqrt{3}$ comes from the weight bound for $Rf_!$ (Theorem 13.11*) applied to the constant sheaf, and $\alpha\beta = 3$ (Example 10.23*) turns it into the lower bound.

The argument of Hasse's theorem works in every dimension. The point-counting formula expresses $\#X_0(\mathbf{F}_{q^m})$ through the traces of F^m, and purity bounds each trace by the dimension of the group times the common absolute value of its eigenvalues.

Corollary 19.10 (Point-count estimates)**.** *Let $X_0/\mathbf{F}_q$ be smooth, proper and geometrically connected of dimension $d \ge 1$, and let b_i be as in Theorem 19.7. Then for every $m \ge 1$,*

$$\left|\#X_0(\mathbf{F}_{q^m}) - (1 + q^{dm})\right| \le \sum_{i=1}^{2d-1} b_i q^{mi/2}.$$

For a smooth proper geometrically connected curve of genus g, this is

$$|\#X_0(\mathbf{F}_{q^m}) - (1 + q^m)| \le 2g q^{m/2}.$$

Proof. Since X_0 is proper, the point-counting formula of Corollary 12.31* reads

$$\#X_0(\mathbf{F}_{q^m}) = \sum_{i=0}^{2d} (-1)^i \operatorname{Tr}(F^m \mid H^i_\ell).$$

The terms in degrees 0 and $2d$ are 1 and q^{dm} by Theorem theorem-weil-conjectures-smooth-proper-schemes. The left side of the estimate is therefore the absolute value of the sum of the remaining terms. In degree i, the trace is a sum of b_i eigenvalues of F^m, each of complex modulus $q^{mi/2}$ by Theorem theorem-weil-conjectures-smooth-proper-schemes. Its absolute value is thus at most $b_i q^{mi/2}$, and the triangle inequality proves the first bound. For a curve, $b_1 = 2g$ by the cohomology calculation of Appendix D*. ∎

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References marked * are to parts of the blueprint this document leaves out:

Corollary 10.13 Degreewise duality over fields, in II Duality and the trace formula (`corollary-degreewise-duality-over-fields`)

Theorem 10.15 The trace isomorphism in top degree, in II Duality and the trace formula (`theorem-trace-isomorphism-top-degree`)

Theorem 10.21 Poincaré duality and Frobenius, in II Duality and the trace formula (`theorem-poincare-duality-frobenius`)

Example 10.23 The running example: duality, in II Duality and the trace formula (`example-running-example-duality`)

Chapter 12 The trace formula, in II Duality and the trace formula (`chap-The-trace-formula`)

Corollary 12.31 Point counting and the zeta function, in II Duality and the trace formula (`corollary-point-counting-zeta-function`)

Example 12.32 The running example: the trace formula, in II Duality and the trace formula (`example-running-example-trace-formula`)

Corollary 12.33 Functional equation, in II Duality and the trace formula (`corollary-functional-equation`)

Theorem 13.11 Weights of $Rf_!$, in III Weights and the Riemann hypothesis (`theorem-weights-rf-shriek`)

Theorem 18.15 Direct image of a mixed sheaf, in III Weights and the Riemann hypothesis (`theorem-direct-image-mixed-sheaf`)

Appendix D Cohomology of torsion sheaves on curves, in Appendices (`chap-Cohomology-of-torsion-sheaves-on-curves`)

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